

# Eric Zhan

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## Education

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### Massachusetts Institute of Technology

Sep 2024 - May 2028

*Bachelor of Science in Computer Science and Engineering (6-3)*

- Relevant coursework: Software Performance Engineering (6.106), Distributed Systems (6.584), Numerical Algorithms for Computing and Machine Learning (6.735), Randomness and Computation (6.542), Design and Analysis of Algorithms (6.122), Software Construction (6.102), Introduction to Machine Learning (6.390).

## Work Experience

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### DRW Trading Group

Jan 2026 - Feb 2026

*Quantitative Trading Intern*

- Analyzed data for options pricing and key macroeconomic indicators in Python.
- Leveraged autoregressive methods and hidden Markov models to improve data adjustment beyond published figures.
- Productionized a data pipeline to provide traders with signal using the higher quality adjusted data.

### Daily Challenge with Po-Shen Loh

Feb 2022 - Jul 2024

*Instructor*

- Effectively taught concepts in math competitions to students using pre-prepared materials from Professor Loh (CMU).
- Collaborated with partner teaching assistants in classes and outreach “livesolve” events.
- Engaged and motivated students to learn; inspired students to enjoy mathematics.

## Research Experience

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### Robust Robotics Group @ MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)

Jun 2026 - Present

*Undergraduate Researcher*

- Designing intelligent exploration algorithms for statistically learning partially observable systems.
- Implementing methods numerically in Python to practically evaluate the effectiveness of different algorithms.
- Optimizing the evaluation codebase by designing custom data structures in C, cutting experiment runtime by 3x.

### FutureTech @ MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)

Jun 2025 - Dec 2025

*Undergraduate Researcher*

- Analyzed and compiled algorithms for intractable computer science problems in literature.
- Automated parsing and validation for 8000 data entries on algorithm complexity classes.
- Integrated data pipelines to fix 100+ errors in previously collected algorithm data.

### MIT Program for Research in Mathematics, Engineering and Science (PRIMES)

Jan 2023 – Jan 2024

*Researcher*

- Improved bounds on an unsolved math conjecture with applications in dynamical systems and number theory.
- Presented my findings at the annual MIT PRIMES conference and at the Brandeis University Dynamics and Number Theory Seminar.
- V. Nekrasov and E. Zhan, *On Nontrivial Winning and Losing Parameters of Schmidt Games*, arXiv e-prints (2024), <https://doi.org/10.48550/arXiv.2401.00614>.

## Awards

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- Regeneron Science Talent Search Scholar (2024), National Merit Scholarship Finalist (2024)
- USA Math Olympiad Bronze Medalist (2023), USA Junior Math Olympiad Winner (2022), Honorable Mention (2021)
- USA Physics Olympiad 2x Silver Medalist (2024, 2023), Bronze Medalist (2022)

## Skills

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- Programming Languages: Python, Typescript, C, Go.
- Web Frameworks: Django, Express, Flask, Next.js, React, TailwindCSS.
- Other: LaTeX, SQLite, AWS, Docker, Linux.

## Projects

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- MapIT, a interactive map of MIT's floor plans to help students navigate campus, serving 1k+ users.
- A full stack website to host the MIT Brass Rat Puzzle Hunt, served 200 teams during the puzzle hunt.
- For additional projects and other information, see <https://ericzhan.me>